

# HYPER-SNN: Towards Energy-efficient Quantized Deep Spiking Neural Networks for Hyperspectral Image Classification

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**Abstract**—Hyperspectral images (HSIs) provide rich spectral-spatial information across a series of contiguous spectral bands. However, the accurate processing of the spectral and spatial correlation between the bands requires the use of energy-expensive 3-D Convolutional Neural Networks (CNNs). To address this challenge, we propose the use of Spiking Neural Networks (SNNs) that are generated from iso-architecture CNNs and trained with quantization-aware gradient descent to optimize their weights, membrane leak, and firing thresholds. During both training and inference, the analog pixel values of a HSI are directly applied to the input layer of the SNN without the need to convert to a spike-train. The reduced latency of our training technique combined with high activation sparsity yields significant improvements in computational efficiency. We evaluate our proposal using three HSI datasets on a 3-D and a 3-D/2-D hybrid convolutional architecture. We achieve overall accuracy, average accuracy, and kappa coefficient of 98.68%, 98.34%, and 98.20% respectively with 5 time steps (inference latency) and 6-bit weight quantization on the Indian Pines dataset. In particular, our models achieved accuracies similar to state-of-the-art (SOTA) with  $\sim 560.6\times$  and  $\sim 44.8\times$  less compute energy on average over three HSI datasets than an iso-architecture full-precision and 6-bit quantized CNN, respectively.

**Index Terms**—hyperspectral images, spiking neural networks, quantization-aware, gradient descent, indian pines

## I. INTRODUCTION

**H**YPERSPECTRAL imaging, which extracts rich spatial-spectral information about the ground surface, has shown immense promise in remote sensing [1]. It is currently used in several applications ranging from geological surveys [2], to the detection of camouflaged vehicles [3]. In hyperspectral images (HSIs), each pixel can be considered as a high-dimensional vector where each entry corresponds to the spectral reflectivity [1] of a particular wavelength. The goal of the classification task is to assign a unique semantic label to each pixel [4].

For HSI classification, several spectral feature-based methods have been proposed, including support vector machine [5], random forest [6], canonical correlation forest [7], and multinomial logistic regression [8]. To improve the accuracy of HSI classification, researchers have integrated spatial features into existing learning methods [9]. Some spectral-spatial methods for classifying HSIs include fusing correlation coefficient and sparse representation [10], Boltzmann entropy-based band selection [11], joint sparse model and discontinuity preserving relaxation [12], and extended morphological profiles [13],

[14]. Some of these methods have also been proposed to exploit the spatial context with various morphological operations for HSI classification. However, these spectral-spatial feature extraction methods rely on hand-designed descriptions, prior information, and empirical hyperparameters [1].

Lately, convolutional neural networks (CNNs) have yielded higher accuracy than some hand-designed features [15]. CNNs have shown promise in multiple applications where visual information processing is required, including image classification [16], object detection [17], semantic segmentation [18], and depth estimation [19]. In particular, CNN-based methods act as an end-to-end feature extractor that consists of a series of hierarchical filtering layers for global optimization. The 2-D CNN stacked autoencoder [1] was the first attempt to extract deep features from its compressed latent space to classify HSIs. Following this work, [20] employed a 2-D CNN model to extract the spatial information in a supervised manner and classify the raw hyperspectral images. The multibranch selective kernel network with attention [21] and pixel-block pair based data augmentation techniques [22] were developed to address the gradient vanishing and overfitting problems. To extract the spatial-spectral features jointly from the raw HSI, researchers proposed a 3-D CNN architecture [23], which achieves even better classification results. Authors in [24], [25], [26] created multiscale spatio-spectral relationships using 3-D CNN and fused the features using a 2-D CNN to extract more robust representation of spectral-spatial information.

However, the performance and success of multi-layer CNNs are generally associated with high power and energy costs [27]. A typical hyperspectral image cube consists of several hundred spectral frequency bands, and hence, classifying these images using traditional CNNs require a large amount of computational power, especially when real time processing is necessary, as in target tracking or identification [28]. The high energy cost and the demand for deployment of HSI sensors in battery-powered edge devices motivates exploring alternative lightweight energy-efficient HSI classification models.

In particular, low-latency spiking neural networks (SNNs) [29] have gained attention because they can be more computationally efficient than CNNs for a variety of applications, including image analysis. To achieve this goal, analog inputs are first encoded into a sequence of spikes using one of a variety of proposed encoding methods, including rate coding [30], [31], direct coding [32], temporal coding [33], rank-order coding [34], phase coding [35], and other exotic coding schemes [36], [37]. Among these, rate and direct coding have

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